MODULE 02 | VERSION 3.0

Planning & Technical Design

Governance-aligned controlled consolidation | 25 October 2026

DOCUMENT CONTROL	APPLICATION
Status	Harmonized business specification; approved prior module workflow retained with controlling Module 7 alignment.
Composition	This V3 addendum and the full previously revised module source are combined in a single PDF.
Priority	This V3 alignment governs in case of conflict. The original module source remains operative where not contradicted.
Central governance	Module 7 is a continuous operational backbone; it is not a gate to be completed after Module 6.
Authority	Documented GM-led delegation and owner-reserved matters replace generic or inconsistent old “Management alone” labels.
Scope of this module	Architectural 2D approval before 3D; structural, mechanical/plumbing, electrical and QS budgeting with independent professional review and controlled versions.

1. Implementation interpretation and source preservation

This document intentionally retains the complete existing revised module specification after this amendment. Do not erase earlier workflow, screenshots, forms, stages or user-approved protections. Configure the shared functions from the V3 Master Architecture and Module 7 specification. The addendum is not permission to skip work or approvals.

2. Shared mandatory company controls

- GOV-01 Governance: The owner retains reserved strategic powers and protected oversight, while the employed General Manager (GM) holds company-wide operational accountability. Owner access to reporting does not authorize routine operational interference.
- GOV-02 Delegation: Department heads, Project Managers and designated officers may authorize decisions only within documented project, function, category, value and risk limits; above-limit or exceptional decisions go to GM. No employee may approve their own originating action.
- GOV-03 Technical independence: Valid safety and professional certifications are performed by competent authorized persons; neither a GM nor a financial waiver can substitute for a mandatory technical/statutory clearance.
- GOV-04 Evidence: Worker, junior designer, client, storekeeper, inspector, supplier and auditor original submissions are preserved with authorship, timestamp, linked versions and tamper-evident history. Review adds new records; it does not overwrite originals.
- GOV-05 Segregation: Requests, verification, approval, payment, receipt and audit must be appropriately independent. Where team size makes perfect separation impractical, a documented compensating review by an independent qualified officer is mandatory.
- GOV-06 Client access: Client portal content is limited to that client/project and expressly published revision; source design files, worker complaints, confidential pay, internal contracts and audit cases remain restricted.

- GOV-07 Controls: Accounts verifies funds/beneficiaries and books actual transactions; QS values cost and forecasts; Contracts verifies commercial entitlements; Compliance oversees obligations; QA checks systemic quality; independent audit observes, investigates and independently verifies closure.
- GOV-08 Exceptions and continuity: Documented financial/administrative exceptions may be decided by the authorized delegated manager or GM; there is no exception to mandatory law, safety, professional status or an already-required client acceptance. GM absence invokes only preapproved and HR-activated acting authority.

3. Updated operating workflow and stage gates

Architectural 2D

Lead Architect assigns juniors, checks versions and obtains evidence of client acceptance of exact 2D layout before any 3D task is released.

Architectural 3D

Assigned junior prepares renders; Lead Architect approves; 3D never substitutes for 2D acceptance.

Engineering chain

Approved internal working set moves to Structural, then Mechanical/Plumbing, then Electrical; each lead delegates and approves discipline scope.

Quantity survey

Lead QS delegates estimators, validates BMS/BQ/budget and prepares summary and detailed controlled estimates.

Proposal handover

Departmentally approved, coordination-checked working revisions form the Module 3 proposal; financial publication requires separately authorized release.

4. Module-specific accountability integration

CONTROL AREA	REQUIRED V3 BEHAVIOR
Design task deadlines	GM/delegated management sets overall target; relevant discipline lead sets junior task deadlines within target.
Professional qualifications	HR stores licences, education, renewals; Compliance monitors expiry; qualified signatories alone certify their regulated work.
Design revisions	Minor correction under authorized scope; major redesign obtains Contracts/QS impact, client change terms and delegated GM/management approval.
Independent QA	Department lead performs mandatory technical acceptance; central QA samples risk, nonconformities and cross-discipline coordination.
Working files	ArchiCAD / editable CAD files internal and versioned; client only specifically published PDF/image outputs.
Project design budget	QS and Accounts independently track design labour/fees and financial actuals; no accidental automatic client proposal change.

5. Conflict resolution register

OLDER / AMBIGUOUS WORDING	V3 INTERPRETATION
“Management” sets overall deadline	Assigned delegated GM/department approval following authority matrix; the owner is not the deadline approver.
Technical “approval” and public “publication”	Separate professional review, commercial acceptance, and delegated management publication permissions.

6. Developer data extensions / shared links

Maintain one stable ProjectId across all modules; preserve exact FileVersionId and client publication authorization, verify role scope server-side and append actor/timestamp/decision history for every state transition.

- DisciplineTask
- WorkingFileVersion/Transmittal
- ApprovalEvidence/Client2DAcceptance
- ProfessionalCredential
- CoordinationIssue
- DesignChangeRegister
- BudgetRevision
- QaFinding/Correction
- AuditEvent

7. Mandatory integration acceptance tests

AT-02-01: Reject any 3D assignment until client acceptance of exact 2D version.

AT-02-02: Reject structural handover without Lead Architect completed release.

AT-02-03: Do not release MEP/QS before predecessor discipline approvals.

AT-02-04: Client downloads cannot expose editable CAD/ArchiCAD files.

AT-02-05: Expired professional credential blocks regulated sign-off but not unrelated work.

AT-02-06: Large redesign requires approved time and commercial impact, preserving original revisions.

8. Required developer references

Read this PDF with 00 Master Architecture, Module 07 Central Administration and 08 Integration & Testing. Additional authoritative user decisions for Modules 1-6 prevail over older illustrative implementation examples where the later decision is more specific. Regulatory obligations must be validated with qualified advisors and current official sources before implementation.

MATHMAG BUILDERS LIMITED

MANAGEMENT SYSTEM

MODULE 02

PLANNING & TECHNICAL DESIGN

Architectural → Structural → Mechanical/Plumbing → Electrical → Quantity Surveying

Plan better, build better.

Organization	Mathmag Builders Limited
Purpose	Revised developer-ready functional specification
Revision	2.0 | 24 October 2026
Approval basis	Confirmed business choices + clearly identified technical implementation recommendations
System scope	Module 02 of 06, on one persistent client/project platform

IMPLEMENTATION RULE

This revision replaces inconsistent stage assumptions in earlier drafts. A confirmed business decision is binding; illustrative fields, example APIs and implementation details must be validated in design review.

01 Purpose, workflow and confirmation status

A five-department controlled design and cost-planning pipeline, with linked evidence, roles, versioned technical releases and client approval gates.

Architectural	Approved Module 1 brief -> 2D -> formal client 2D approval -> 3D -> formal client 3D approval and architectural package release.
Structural	Architectural released native working files -> delegated analysis, design and detailing -> Lead Structural Engineer approval.
Mechanical / Plumbing	Structural release -> delegated mechanical/plumbing coordination and technical review by Lead Mechanical Engineer.
Electrical	Mechanical release -> delegated electrical design and Lead Electrical Engineer technical approval.
Quantity Surveying	Coordinated four-discipline release -> Lead QS assignment, quantity takeoff and approved summary/detailed cost outputs for Module 3.

MANDATORY CONTROL

2D client approval is a mandatory prerequisite before 3D work begins. The departments are ordered as confirmed; proposed deliverable details within new MEP units must be project-scope configurable.

02 End-to-end technical design workflow

01	Module 1 Release	Lead Architect receives approved brief and assigns 2D junior.
02	2D Architectural	Junior drafts and Lead Architect approves client-facing version; client formally accepts the specific layout.
03	3D Architectural	Assigned 3D author creates renderings from approved 2D; Lead Architect reviews and client formally approves version.
04	Structural	Lead Structural Engineer delegates and approves structural analysis, design and detailing.
05	Mechanical / Plumbing	Lead Mechanical Engineer delegates, checks cross-discipline interfaces and releases.
06	Electrical	Lead Electrical Engineer delegates, checks technical and spatial coordination and releases.
07	QS / Budgeting	Lead QS assigns estimators, ensures source versions consistent and technically approves both budget views.
08	Module 3 Handoff	Lock coordinated package and cost estimate snapshot for proposal presentation.

03 Departmental owners and segregation of duties

Lead Architect	Delegates 2D and 3D; sets task deadlines; reviews technical output; controls architectural client publication; releases package after approvals.
Junior Architect / Renderer	Produces task-specific outputs, submits drafts/source files and evidence, responds to review comments; cannot self-approve.
Lead Structural Engineer	Owns structural design and final technical signoff, junior allocations, technical deadlines and release to Mechanical.
Lead Mechanical Engineer	Owns plumbing/mechanical review, junior assignments and release to Electrical.
Lead Electrical Engineer	Owns electrical review, junior assignments and release to QS.
Lead Quantity Surveyor	Owns estimating allocations, technical budget accuracy, version checks and cost-package release.
Client	Reviews published PDFs/images; approves 2D and 3D versions through portal or signed evidence; no access to internal source.
Management	Sets overall design deadlines, controls business scope and major changes, views all task/authorship evidence and dependencies.

04 Shared assignment, task and review engine

• Every technical task must have project, discipline, package revision, assignee, originating instruction, planned deadline, submission, reviewer and status.

• System supports draft, submitted for review, returned with comments, technically approved, released and superseded states; approval and release are separately logged.

• The reviewer may create numbered correction comments; the author submits a new working version; earlier versions remain accessible to appropriate staff.

• Management controls overall design/project target; department leads allocate own junior tasks and set subordinate deadlines.

• Individual professionals have attributable submission histories and task-quality/turnaround records. Department leads cannot silently erase or alter junior-originated work evidence; management retains visibility.

• Management can assess employee performance using timely reporting, completeness, verified technical quality and rework, with context for client changes or missing inputs.

05 Architectural 2D phase - client approval gate

Lead Architect and junior workflow

• Receive the approved site report and brief; assign an authorized junior 2D task with scope and deadline.

• Junior prepares drawings/layout variants, stores editable ArchiCAD project and controlled PDF/image proposals in internal workspace.

• Lead Architect technically reviews the chosen 2D issue; return comments or authorize a specific client-facing revision.

Client review and formal evidence

• Support portal review or physical/virtual design meeting. Attach meeting notes to the exact shown version.

• Accept portal approval or uploaded client-signed approval form bound to a specific 2D version and authorized client identity.

• No 3D task activation until a valid formal 2D acceptance exists. Mere viewing or verbal discussion is not approval.

• Scope-controlled minor changes can cycle normally; substantial redesign requests require management authorization and may entail fees under approved terms.

06 Architectural 3D phase and release

• Lead Architect may assign 2D and 3D to one junior or different juniors, preserving separate task ownership and deadlines.

• 3D must reference the exact client-approved 2D drawing revision; do not produce an unapproved alternative layout as the client deliverable.

• Submit still images/renderings and any contracted visualization outputs for lead review before selective client publication.

• Client comments and revisions must be version-linked; formal 3D acceptance via portal or signed form is retained.

• The architectural internal handover includes native ArchiCAD/source files and locked client-approved planning PDFs/images; native files never appear in the client portal.

• If approved 2D is later materially changed, trigger dependent 3D, structural, MEP and QS revalidation as needed.

07 Structural engineering submodule

• Lead Structural Engineer accepts the approved architectural ArchiCAD working package internally and delegates specific analysis/design/detailing tasks to juniors.

• Structural packages contain applicable analysis assumptions, design calculations, foundation/ground floor/upper structure details, connections, notes, schedules and coordination issues.

• Lead Structural Engineer reviews calculation/drawing compatibility and is sole discipline technical release owner; professional certification remains with appropriately authorized persons.

• Management sets overall structural deadline; lead sets junior task deadlines and returns incomplete tasks for correction.

• Do not pass structural incomplete or unreviewed packages to costing; release the signed-off technical reference package onward to Mechanical / Plumbing.

08 Mechanical and plumbing submodule

• Upon structural release, the Lead Mechanical Engineer receives the compatible architectural and structural source references.

• Lead delegates plumbing/water supply, drainage, sanitary fittings, equipment, ventilation/mechanical systems as relevant to contracted scope and project type.

• Junior deliverables may include PDFs, working files, system sizing, schedules and inspection-ready drawings where required; exact content remains configurable.

• Lead records discipline checks, clash/penetration coordination, corrections and a final technical package release to Electrical.

• Changes to structure/architecture must create cross-discipline coordination issues and revalidate dependent cost items.

09 Electrical submodule

• Lead Electrical Engineer receives the latest compatible approved Architecture, Structural and Mechanical/Plumbing packages.

• Tasks may cover lighting, power, load distribution, circuits, low-current services, equipment points and relevant schedules in accordance with scope.

• Lead reviews technical design, electrical safety matters, coordination with plumbing and structural elements and version completeness.

• Approved Electrical release is made available to QS; any corrections to upstream disciplines trigger controlled dependent revisions.

10 Quantity surveying and budgeting submodule

• Lead QS takes control only after required four discipline package approvals and assigns quantity takeoff and estimating tasks to estimators.

• Quantities must reference exact approved drawing revisions, units, measurement assumptions and scope exclusions; inconsistencies become coordination questions.

• Prepare both summary budget and detailed budget. Lead QS gives technical approval before Module 3 publication consideration.

• Detailed budget remains permission controlled; client-facing summary or detailed view is an explicit management publishing choice in Module 3.

• Changes with measurable cost consequences require re-estimation; preserve pricing assumptions, source date and estimate version.

• If a client only contracts selected design services, Module 2 still records the appropriate deliverable package and honest exclusions, without implying every discipline was commissioned.

11 Coordination issues and controlled revisions

01	Issue Raised	Any downstream team logs clash, missing information, inconsistent dimension or scope change with drawings/screenshots.
02	Owner Assigned	Affected discipline head accepts responsibility and sets resolution task and deadline.
03	Corrected Source Released	Department head approves new version; old issue set remains archived.
04	Impacted Departments Flagged	System marks all affected downstream drawings/budgets as Revalidation Required.
05	Revalidation Completed	Impacted leads acknowledge corrected documents and regenerate dependent outputs.
06	QS / Client Impact Checked	Cost/time implications escalate to management and to client reacceptance where necessary.

MANDATORY CONTROL

Do not silently overwrite a client-accepted drawing, pricing proposal or professional approval. Revision impact must be evaluated through Modules 3, 4, 5 and 6 when relevant.

12 File management, visibility and milestones

Internal files	Authorized technical teams may share images, PDFs and ArchiCAD/native engineering files; restrict by project, discipline and assignment.
Client files	Only selected image/PDF outputs; no native ArchiCAD models, internal calculations or staff review comments by default.
Document version	Maintain author, uploader, file hash, creation date, review, approval, release, supersession and dependency manifest.
Lead deadlines	Management sets overall planned design delivery; Lead Architect sets 2D/3D assignments; Structural Lead sets structural junior targets.
MEP and QS deadlines	Provide configurable lead/junior task dates rather than invent a non-confirmed fixed company policy.
Evidence integrity	Junior submission and correction history remain independently accessible to management; no staff may conceal a subordinate original submission.
Client approvals	Formal 2D and 3D approvals tie to version and client/representative identity.

13 Shared rules carried across all six modules

One lifecycle record	Maintain one project ID from confirmed site-visit booking through design, approvals, contracts, construction and post-handover. Never create a disconnected project just to open a new module.
Role separation	User role, project assignment and explicit approval authority determine every permitted action. Technical sign-off, financial validation, management approval and client acceptance are distinct records.
Evidence integrity	Preserve original submissions, attachments, timestamps, authors and immutable decision histories. Corrections create new records or versions; no silent overwriting.
Selective client access	Client accounts can access only their own projects and explicitly authorized files or updates. Internal working/source files, staff performance data and financial notes are never published by default.
Conditional service scope	Clients may purchase design only, printing only, approvals, technical supervision, labour-only building or full construction. Modules not contracted are not mandatory project steps.
Management-controlled exceptions	Any permitted exception requires identified management authorization, reason, date and retained evidence. Mandatory legal/technical clearances cannot be overridden.
Security and audit	Require authenticated accounts, least privilege, encrypted transport/storage, file scanning, access logs, project-scoped permissions, backups and recoverability.

14 Screens and user journeys

Department Pipeline	Five-stage dependency board with technical gates and package release history.
Lead Architect Review	2D/3D tasks, versions, review comments, client review/publish queue and approvals.
Junior Technical Workspace	Assigned workload, file uploads, review rounds, deadlines and handover checklists.
Client Design Review	2D and 3D authorized PDF/image viewers, comments and signed/portal acceptance.
Structural Workspace	Native file handover, calculation attachments, junior tasks and lead approvals.
Mechanical/Electrical Workspaces	Coordination references, component schedules, correction cycle and lead release.
QS Workspace	Versioned source drawings, quantities, rates, summary/detailed budget and approval.
Management Analytics	Department turnaround, active rework, original authorship trail and schedule forecasts.

15 Core data objects and key relationships

Logical entities for developer schema design. Names below are examples rather than prescribed database table names.

DesignPackage / DisciplineRelease	Project, discipline, source manifest, issue version, released by/at, downstream dependency.
DesignTask / ReviewComment	Assignee, target, date, original upload, comments, correction rounds, reviewer, status.
ClientDesignApproval	Client identity, 2D/3D phase, exact document revision, portal or signed evidence, date.
NativeFile / ClientFile	Access classification, file hash, approved presentation version and upload lineage.
CoordinationIssue	Owner discipline, affected documents, severity, action/revalidation list, due date.
CostEstimate / EstimateLine	QS owner, revision, quantity, unit, rate, amount, source version, summary/detail rights.

DesignRevision / ImpactCase	Origin, minor/major classification, manager decision, client acceptance dependencies.
AuditEvent / PublicationGrant	All read/write/publish decisions and their immutable approvals.

16 Illustrative service and event contracts

Suggested API/service boundaries for planning. Authentication, validation, permission checks, idempotency and audit metadata apply to all write operations.

Create/submit design task	POST /projects/{id}/design-tasks; POST /design-tasks/{id}/submit.
Review and release	POST /design-packages/{id}/review; POST /design-packages/{id}/release.
Record client approval	POST /projects/{id}/design-approvals with version and acceptance evidence.
Publish design	POST /projects/{id}/client-publications, approved image/PDF only.
Log coordination issue	POST /projects/{id}/coordination-issues; POST /issues/{id}/resolve.
Events	2DApprovedByClient, 3DApprovedByClient, StructuralReleased, MechanicalReleased, ElectricalReleased, QSApproved, Module3Ready.

17 User acceptance tests and release criteria

• TEST 01 - Reject creation of 3D execution task without valid client 2D approval on exact version.

• TEST 02 - Accept portal and signed-form approvals, retaining identity and exact design revision.

• TEST 03 - Prevent client from downloading native ArchiCAD files or unapproved internal drafts.

• TEST 04 - Lead Structural Engineer approves structural release; Electrical and QS cannot skip mechanical dependency without explicit scope logic.

• TEST 05 - Show original junior submission and supervisor/lead correction separately to management.

• TEST 06 - Force downstream revalidation and costing review when structural source file changes.

• TEST 07 - QS budget cannot pass Module 3 without Lead QS technical approval.

• TEST 08 - Allow 2D and 3D to be assigned to same or different architects, with independent task deadlines and audit.